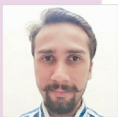


MACHINE LEARNING BASICS

For Newbies



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When we start the journey of life as new-born babies, we inherit the characteristics of our parents. We do not know what to do and when to do what. As we grow up, our parents and elders teach us how to walk, talk and take various decisions in our lives and, as time passes, we gain experience and knowledge. Finally, we start taking our own decisions based on our learning and experience.

Similarly, when we write any code to make a system do any work, the system only does what we ask it to do—it cannot think or take any extra decisions on its own, nor perform actions on that basis. Machine learning teaches the system to learn and take decisions when exposed to a new set of data on the basis of the experience it gains while performing different actions.

Machine learning is an emerging technology that is widely being implemented across all types of industries. Google's self-driving cars, flying drones, anomaly detection and Big Data processing are among the recent examples.

Machine learning is a type of artificial intelligence that provides computers with the ability to learn without being explicitly programmed. It uses pattern recognition and computational learning theory to study and develop algorithms (which can learn from the sets of available data), on the basis of which it takes decisions. These algorithms work by building a model (such as predictive model or neural network model) from sample inputs in order to take data-driven decisions. These models help in developing decision trees, using which the system takes its decision.

Machine learning makes use of mathematical optimisation to deliver different theories, methods and application domains for a specific field. It uses the data-mining technique to perform exploratory data

analysis over a set of data in order to make predictions. This is referred to as unsupervised learning.

Machine learning helps data scientists, engineers, researchers and analysts take a reliable decision by uncovering the hidden insights acquired through the analysis of historical trends in data.

Types of machine learning

Tasks performed using machine learning are classified broadly into three categories, based on the nature of the learning signal available to a learning system (that helps take decisions).

Supervised learning. This is a type of machine learning in which the system is presented with a set of labelled training data (inputs and their corresponding set of outputs). Now, it is the task of supervised machine learning to predict a new set of outputs for a given new set of inputs by learning or finding out a general rule or pattern that maps the given set of inputs to their corresponding outputs. The pattern or rule that helps in predicting output is generally denoted by a specific function. Supervised learning is further classified as regression and classification problems, on the basis of the methodology that is implemented to find a specific pattern.

Unsupervised learning. This machine learning technique is implemented when there is only a set of inputs available with the system, with no corresponding outputs. Now, it is left to the system to learn and identify the pattern or rule governing the available inputs by using unsupervised learning and, further, that hypothesis or rule is used to find the output for the given set of inputs.

There can be many possible hypotheses, but the optimal one is considered for finding the output. Again, unsupervised learning technique is further classified as k means and hierarchical clustering problems, on the basis of the different techniques used to find the final hypothesis.

Fig. 1: Classification of machine learning techniques (Image courtesy: www.analyticsvidhya.com)

